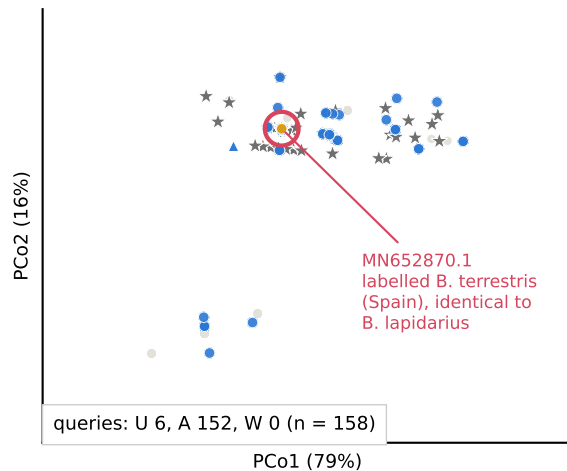
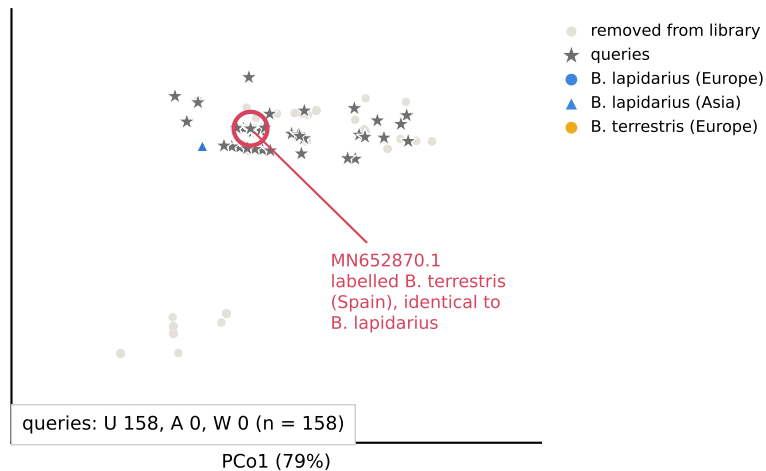


a *B. lapidarius* queries (Central Europe)
block removed (LOBO)

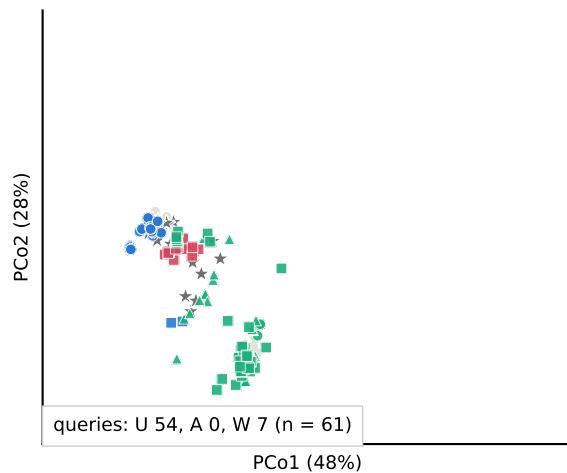


b same queries
continent removed (LOCO)

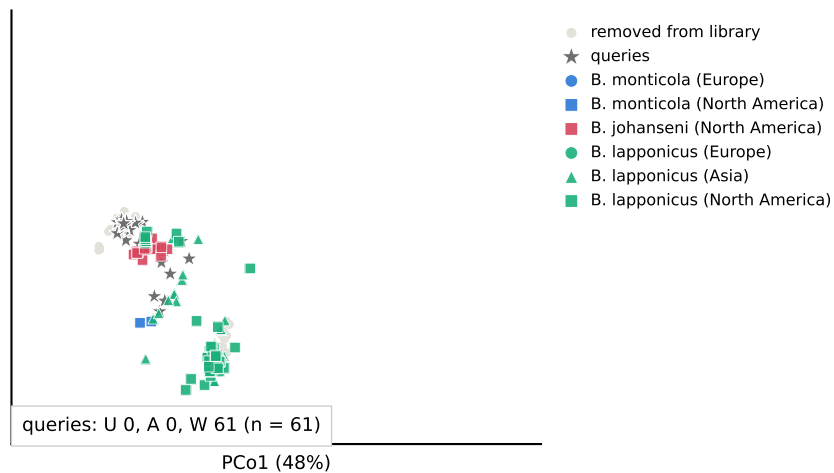


- removed from library
- ★ queries
- *B. lapidarius* (Europe)
- ▲ *B. lapidarius* (Asia)
- *B. terrestris* (Europe)

c *B. monticola* queries (Northern Europe)
block removed (LOBO)



d same queries
continent removed (LOCO)



- removed from library
- ★ queries
- *B. monticola* (Europe)
- *B. monticola* (North America)
- *B. johanseni* (North America)
- *B. lapponicus* (Europe)
- ▲ *B. lapponicus* (Asia)
- *B. lapponicus* (North America)

PCoA of core COI p-distances (axes 1–2 explain 95% (a, b) and 76% (c, d) of positive variation; view zoomed to exclude a few distant records). hapshared library. In (a, b) only *B. lapidarius* and the single *B. terrestris*-labelled record MN652870.1 are shown; every other *B. terrestris* record is $\geq 0.4\%$ from any *B. lapidarius*. Competitor removal (a→b): a single identical, differently labelled record ties with the queries until the continent is removed. Conspecific loss (c→d): without European *B. monticola* all 61 queries are misidentified, 52 of them as the North American *B. johanseni*.